



The Dexter plague

Dexter Malware makes psychokiller into cyberthief. I see plastic cards in your future. <http://dgit.in/1gNhdoH>



Real world DND

Not unlike their online shopping counterparts, retail stores in the U.S are tracking their customer's every move. Here's how <http://dgit.in/1gCBZtE>

Keyboards 101

Buying something as simple as a keyboard for your PC can be quite a task with all those options out there. Let's help you cut that long list down, shall we?

USB or PS/2

Most motherboards come with a PS/2 port so it really doesn't matter if you get a PS/2 or USB keyboard unless you want the portability that comes with it. Keyboards don't need any significant bandwidth so use a USB 2.0 port.

Key Roll Over

KRO is the ability of a keyboard to distinguish all the keys that are simultaneously pressed.

- 3KRO - can detect 3 keys
- 6KRO - can detect 6 keys
- nKRO - can detect any number of keys

Ghosting

The keyboard circuitry is a mesh of multiple electrodes. Pressing a key triggers a combination in the mesh; pressing two triggers another. Certain combinations share the same signals so there is a chance that a key that wasn't pressed might be detected purely because of the similar combination in the mesh. This is called ghosting. Good motherboards have a modified mesh so that multiple combinations don't share the same signal.

Layouts

The way keys are positioned on the keyboard is according to a pattern that has been carried forth from the typewriters. Current technology has overcome the hindrances of the mechanical typewriter and thus we now have multiple keyboard layouts. Certain countries prefer a different positioning of some keys. Also, the language of the keyboard has a lot to do with the layout as well. The QWERTY layout is the most common, but there are other interesting layouts like the DVORAK - the most commonly used letters are placed in the centre row (ASDFGHJKL). This improves typing efficiency by reducing the number of mistakes made.

Unusual Keyboard Types

Foldable

Made out of rubber or silicon, these keyboards can be rolled up and carried easily. Some aren't flexible but they can be dismantled.



Optical/Laser

These keyboards are quite exotic, they have a flat surface onto which the entire keyboard pattern is illuminated by light. Placing your finger over a key reflects light back into the keyboard and triggers a keypress.

Projection

Similar to optical keyboards with a twist, projection keyboards project the entire keyboard pattern onto any surface that reflects a sufficient bit of the projected light. Placing any reflective object over a key reflects light which is captured by a camera.

Features

Mechanical keys

These are durable keys whose trigger mechanism is altered to provide tactile feedback for those who prefer an audio chatter as they type.

Cherry MX Keyboards: Features			
	Tactile feedback	Click noise	Actuation force
Cherry MX Black	No	No	60 cN
Cherry MX Red	No	No	45 cN
Cherry MX Brown	Yes	No	NA
Cherry MX Blue	Yes	Yes	50 cN
Cherry MX Clear	Yes	No	65 cN
Cherry MX Grey	NA	NA	80 cN
Cherry MX Green	Yes	Yes	80 cN
Cherry MX White	Yes	Slight	80 cN
Cherry MX Super Black	No	No	150 cN
Cherry MX Dark Grey	No	No	80 cN
Cherry MX Lock	NA	NA	NA

The Cherry MX series which is popularly used in most keyboards comes in multiple variants. Each variant suited for specific purpose.

Macros

Macro keys are programmable buttons which can trigger a set of commands or multiple key presses in a sequence. So if you want just one key to open an Explorer window, a browser window, a calculator and the notepad with just one key.

Backlighting

Backlighting is achieved when the entire key is made with an infusion of opaque and translucent plastic. The translucent part is then painted over with opaque paint to leave out just the key pattern. So when an LED is lit up beneath the keyboard only the character shows up illuminated. Certain keyboards allow for patterns in backlighting so you can have certain keys (WASD) light up keeping the others dark.

Cable modifications

Sometimes the keyboard cable is covered with a protective cloth sheath for added durability. Also keyboards feature routing paths along the bottom to channel the cable out of whichever side you want to. Now-a-days keyboards sometimes feature removable cables so you can replace the cable in case a mouse chews through it. Gold plating has no effect on performance other than the fact that it doesn't oxidize.